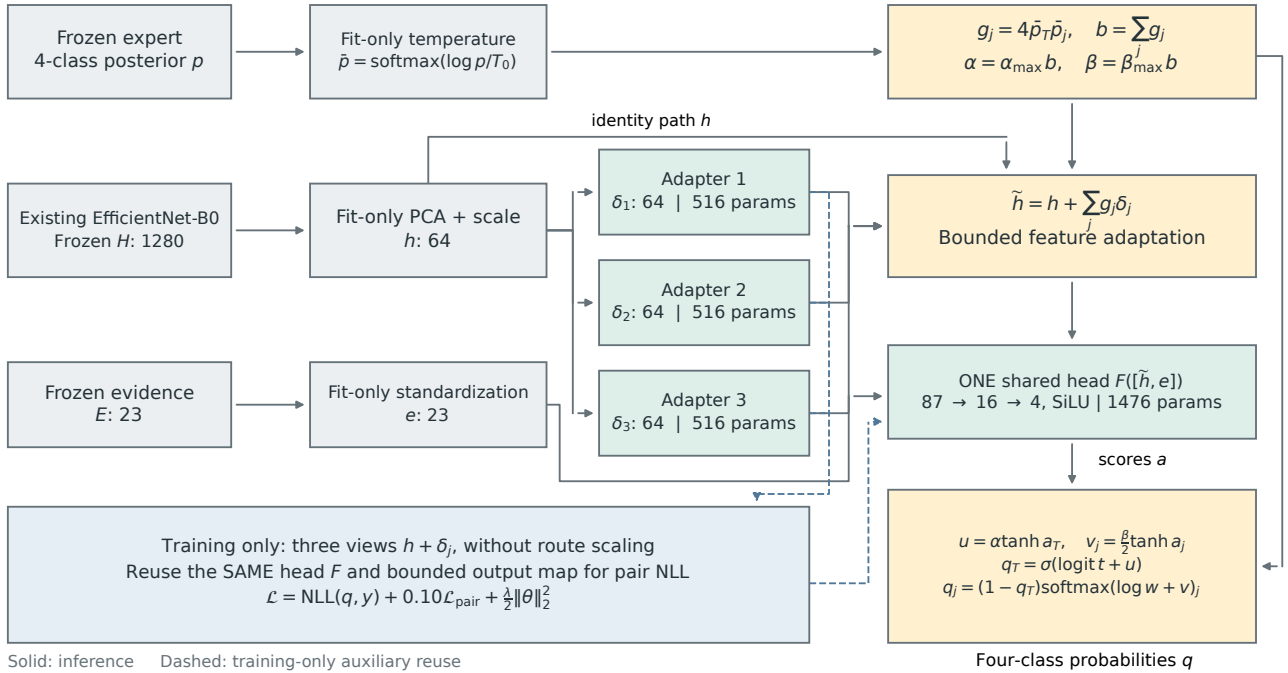


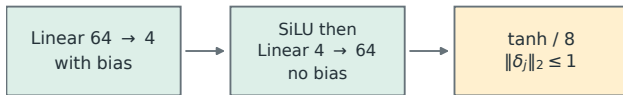
LC-RFA-B: lightweight role-conditioned feature adaptation

Implemented architecture | Effectiveness awaits real-data evaluation

a



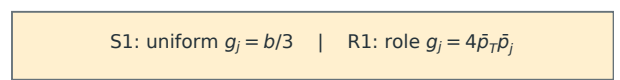
b Each bounded adapter



$$\delta_j = \frac{1}{\sqrt{64}} \tanh[U_j \text{SiLU}(V_j h + b_j)]$$

$$\text{Parameters: } (64 \times 4 + 4) + (4 \times 64) = 516$$

c Matched routing comparison



Same three adapters, shared head, pair loss and budget.

$$\text{R1 trainable parameters: } 3 \times 516 + 1476 = 3024$$

$t = \bar{p}_T, w_j = \bar{p}_j/(1 - t), j \in \{\text{Ti, G, M}\}; T$: target proxy; Ti/G/M: three non-target roles.

Frozen image features and evidence are inputs. Labels enter the training loss only. Bounds do not guarantee accuracy or recall.